

Face Recognition System

Introduction

Face recognition is a computer vision technique used to identify or verify a person using facial features.

In this project, we built an AI-based face recognition system using Python, CNN, and OpenCV that can recognize known persons, detect unknown persons, compare faces, and retrieve person details from a database.

Dataset Used

Two datasets were used.

1. AT&T Face Dataset
 - Contains images of 40 different people.
 - Each person has multiple face images.
 - Used for training the model.
2. Custom Dataset
 - We created our own dataset with images of friends.
 - Each folder represents one person with multiple images.

Image Preprocessing

Before training the model, images were preprocessed.

Steps performed:

- Convert images to grayscale
- Resize images to 64×64 pixels
- Normalize pixel values

Preprocessing helps improve model accuracy and training performance.

Data Splitting

Split dataset into training and testing data

Example:

Training Data – 80%

Testing Data – 20%

Neural Network Model (CNN)

We used a Convolutional Neural Network (CNN) for face recognition.

CNN structure: Convolutional Neural Networks (CNNs or ConvNets) are specialized deep learning models designed to automatically learn hierarchical spatial features from data, primarily images, using convolutional, pooling, and fully connected layers.

CNN automatically learns facial patterns and features from images.

Model Training

The model was trained using the Backpropagation algorithm.

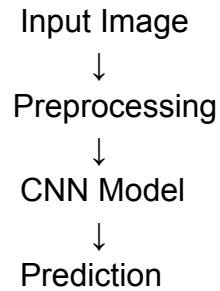
Process:

1. Input image is passed through the CNN.
2. The model predicts a label.
3. Error is calculated.
4. Weights are updated using gradient descent.

Model accuracy achieved: $\approx 85\text{--}90\%$

Face Recognition

When a new image is given:



Output example:

Predicted Person: C3

Confidence: 0.91

Unknown Face Detection

If the model sees a person not in the dataset, we use a confidence threshold.

Example:

If confidence > 0.7 → Known person

If confidence < 0.7 → Unknown person

Face Verification

The system can also compare two images to check if they belong to the same person.

Steps:

Image 1 → Feature Extraction

Image 2 → Feature Extraction

↓

Calculate similarity

If similarity is high → Same person

If similarity is low → Different person

Cosine Similarity

To compare faces, we used Cosine Similarity.

Formula: $\cos(\theta) = (A \cdot B) / (||A|| \ ||B||)$

- Value close to 1 → Same person
- Value close to 0 → Different person

Person Database

A database was created to store person details.

Example:

```
people_db = {  
  "C1": {  
    "name": "Akanksha",  
    "age": 20,  
    "email": "...",  
    "mobile": "..."  
  }  
}
```

The system can search using:

- Person ID
- Email
- Mobile number

and retrieve person details and image.

Most Important Algorithms Used

1. Convolutional Neural Network (CNN) – for face recognition.
2. Backpropagation – for training the neural network.
3. Cosine Similarity – for comparing two faces.
4. Label Encoding – for converting names into numerical labels.

Final System Workflow

